

YUBI Gripper_DYNAMIXEL_ver.2 Assembly Procedure

[Assembly State]

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(1) Installation of components onto UPPER PLATE

- Install inserts
- Install locating pins

(2) Installation of components onto CASE

- Install inserts

(3) Installation of components onto DYNAMIXEL

- Install supplied wheel
- Install GEAR SHAFT

(4) Installation of components onto FINGER ATTACHMENT_R

- Install FINGER PAD
- Install RUBBER SHEET

(5) Installation of components onto FINGER ATTACHMENT_L

- Install FINGER PAD
- Install RUBBER SHEET

(6) Installation of DYNAMIXEL

- Remove bottom bolts
- Install BRACKET_DYNAMIXEL
- Install CASE

(7) Installation of FINGER ASSY_R (Right Finger, Step (4))

- Install SPUR GEAR
- Install FINGER ASSY_R

(8) Installation of FINGER ASSY_L (Left Finger, Step (5))

- Install SPUR GEAR
- Install FINGER ASSY_L

(9) Installation of UPPER PLATE

- Install bearings
- Install UPPER PLATE

(10) Installation of FLAPs

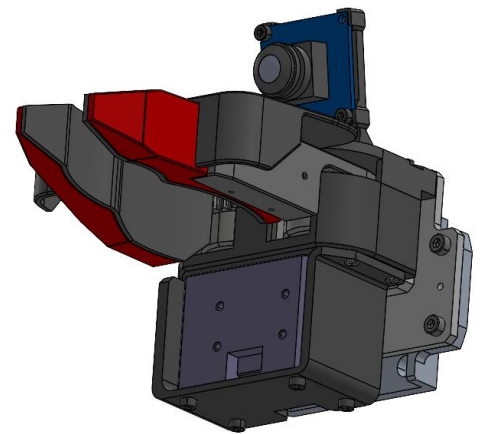
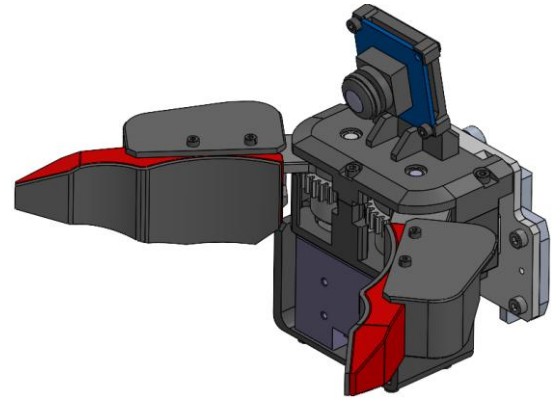
- Install FLAP_R
- Install FLAP_L

(11) Installation of CAMERA

- Install CAMERA

(12) Installation of YUBI Gripper_DYNAMIXEL onto UR5e

- Install BRACKET_GRIPPER
- Install UR5e_FLANGE onto UR5e
- Install YUBI Gripper_DYNAMIXEL onto UR5e

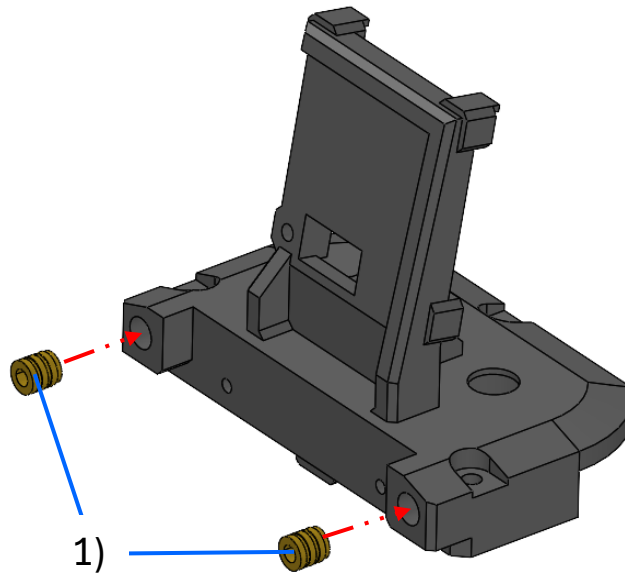


(1) Installation of components onto UPPER PLATE

- Install inserts

Install the following inserts into UPPER PLATE.

1) SB-304540: 2 pcs

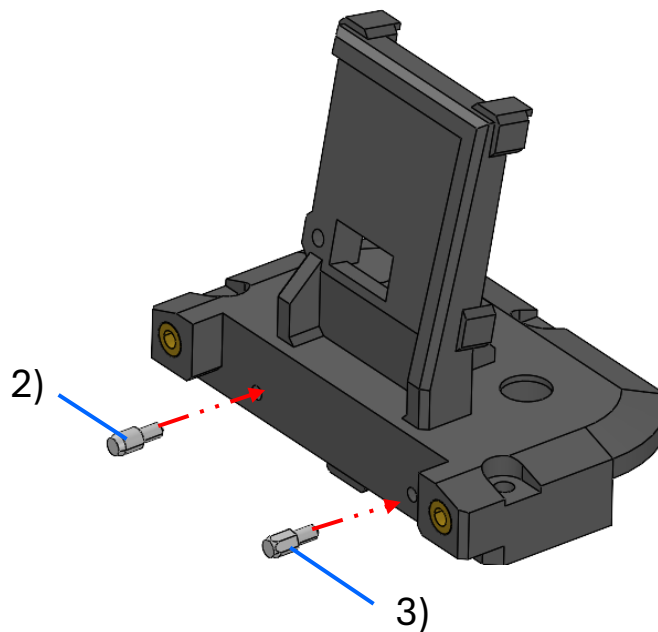


- Install locating pins

Install the following locating pins into UPPER PLATE.

2) JPBPB2-3: 1 pc

3) JPDPB2-3: 1 pc



(2) Installation of components onto CASE

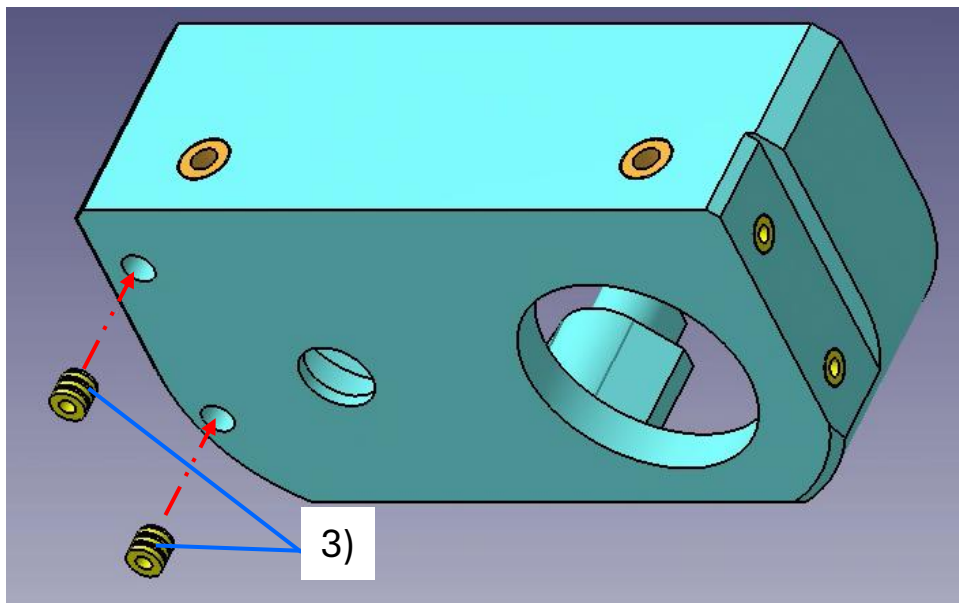
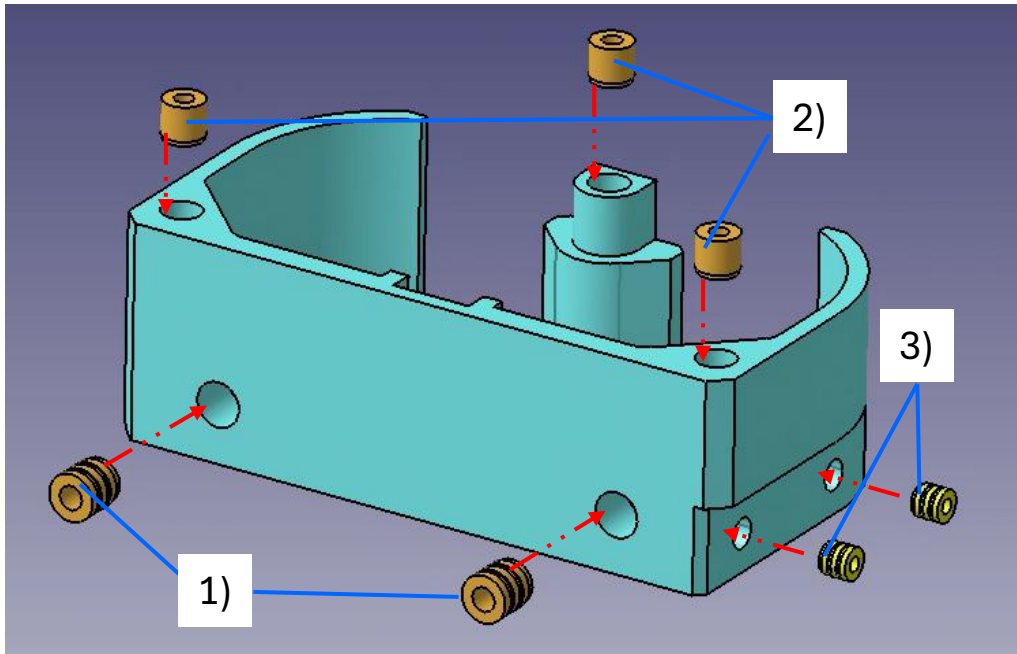
- Install inserts

Install the following inserts into CASE

1) SB-304550: 2 pcs

2) SB-264040: 2 pcs

3) SB-203030: 4 pcs



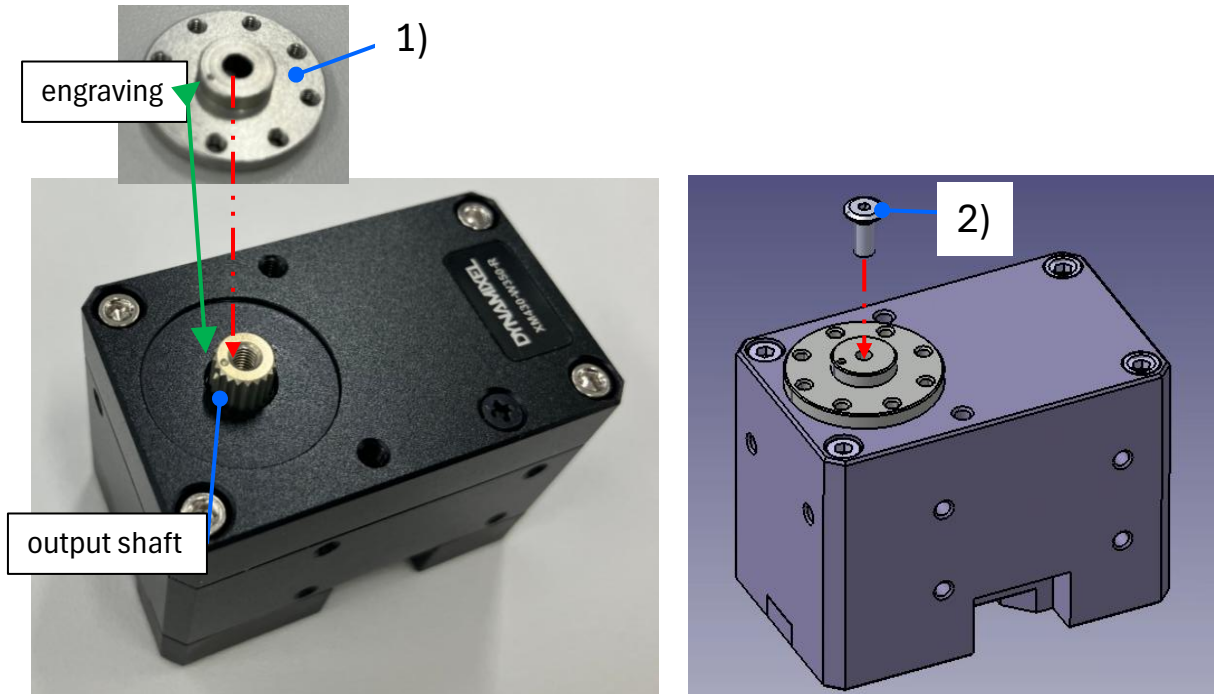
(3) Installation of components onto DYNAMIXEL

- Install supplied wheel

1) Install supplied wheel onto the output shaft of DYNAMIXEL XM430-W350-R.

Align the position of the engraved marks before installation.

2) Secure the wheel with CBSTNR2.5-6. (Torque: $0.65 \text{ N}\cdot\text{m}$)



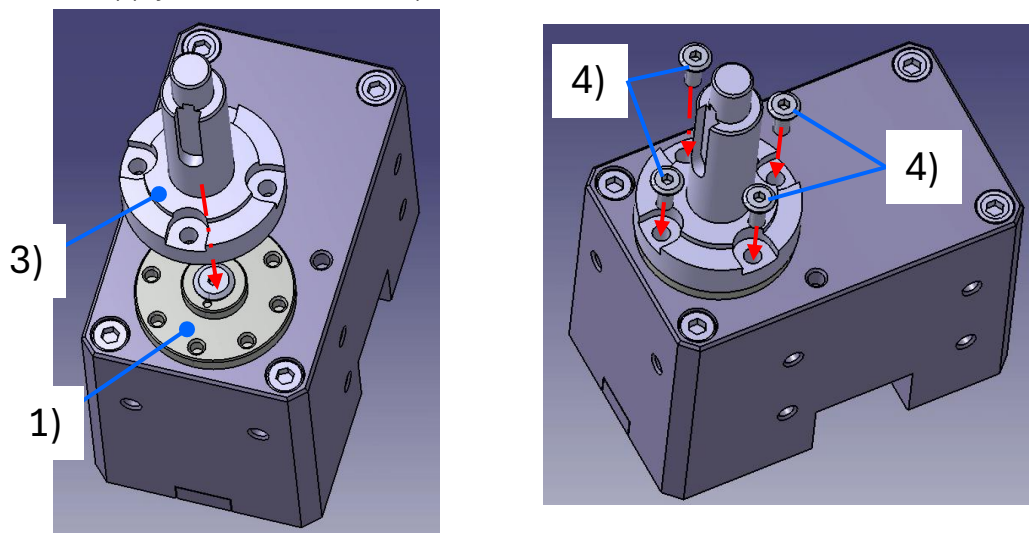
- Install GEAR SHAFT

3) Install GEAR SHAFT onto the wheel.

Align the keyway of GEAR SHAFT with the alignment mark on the wheel.

4) Secure GEAR SHAFT with CBSTNR2-5 (4 pcs).

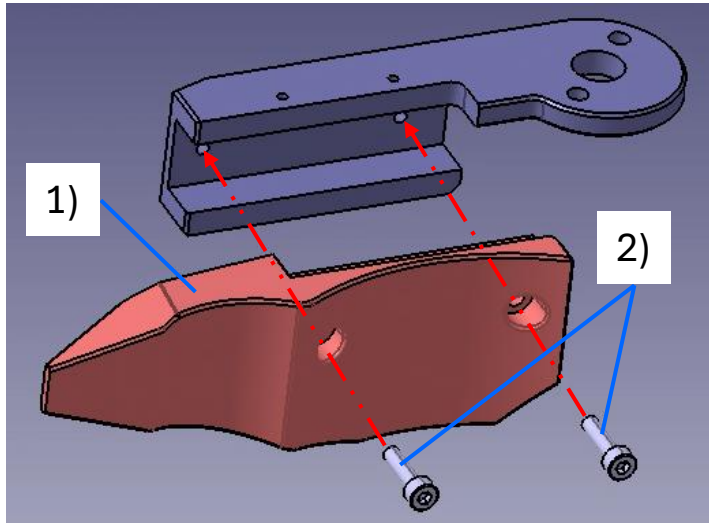
(Apply threadlocker, Torque: $0.315 \text{ N}\cdot\text{m}$)



(4) Installation of components onto FINGER ATTACHMENT_R

- Install FINGER PAD

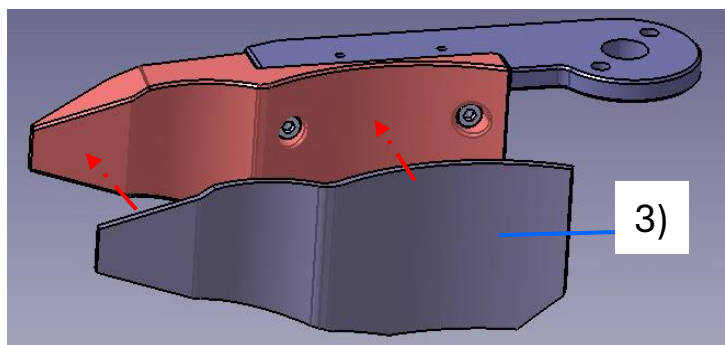
- 1) Install FINGER PAD_t30_R or FINGER PAD_t20 onto FINGER ATTACHMENT_R.
- 2) Secure FINGER PAD with CB2.5-10 (2 pcs). (Torque: 0.36 N•m)



- Install RUBBER SHEET

- 3) Attach RUBBER SHEET to FINGER PAD using suitable adhesive.

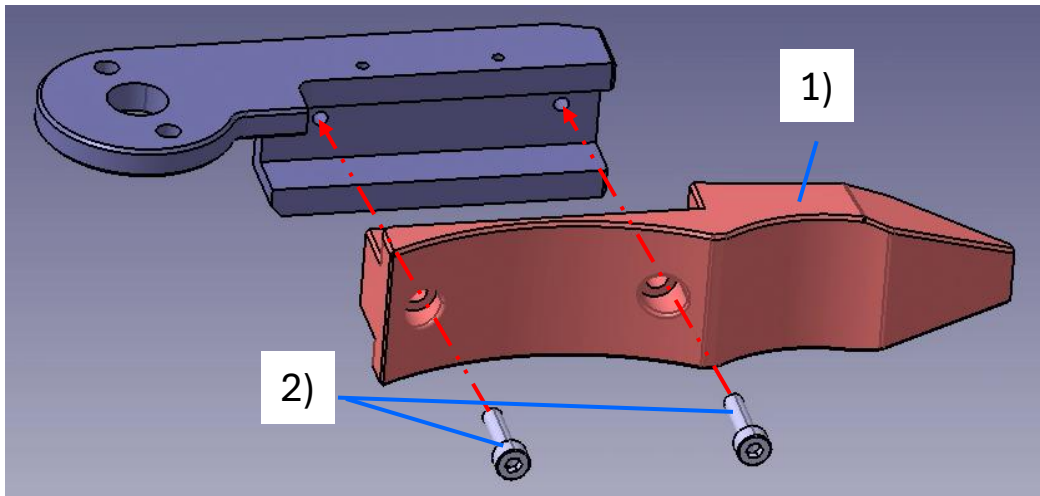
The size of RUBBER SHEET may be selected based on the gripping location and required area.



(5) Installation of components onto FINGER ATTACHMENT_L

- Install FINGER PAD

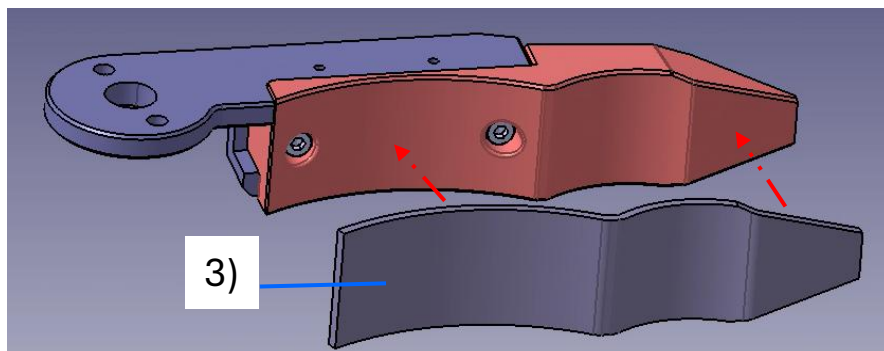
- 1) Install FINGER PAD_t30_L or FINGER PAD_t20 onto FINGER ATTACHMENT_L.
- 2) Secure FINGER PAD with CB2.5-10 (2 pcs). (Torque: 0.36 N·m)



- Install RUBBER SHEET

- 3) Attach RUBBER SHEET to FINGER PAD using suitable adhesive.

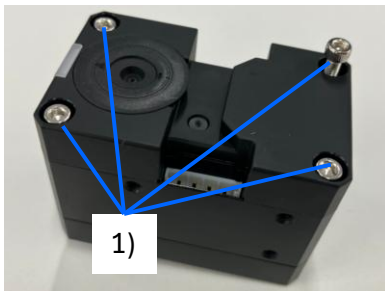
The size of RUBBER SHEET may be selected based on the gripping location and required area.



(6) Installation of DYNAMIXEL

- Remove bottom bolts

1) Remove the four bolts on the bottom side.

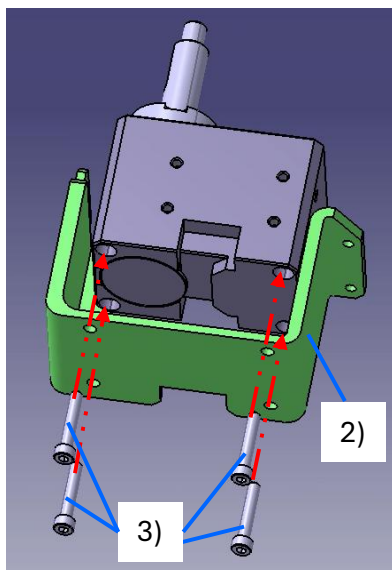


- Install BRACKET_DYNAMIXEL

2) Install BRACKET_DYNAMIXEL onto DYNAMIXEL + GEAR SHAFT ASSY (3).

Be careful not to loosen or detach the DYNAMIXEL bottom cover during assembly.

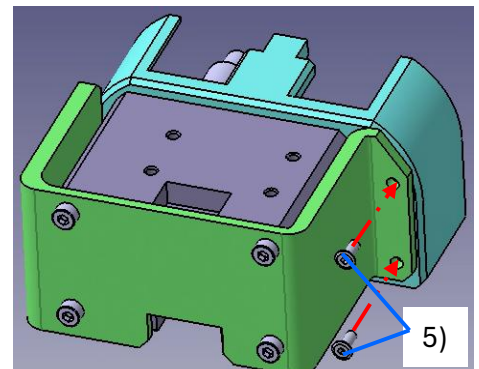
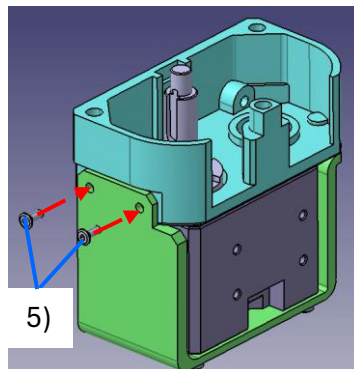
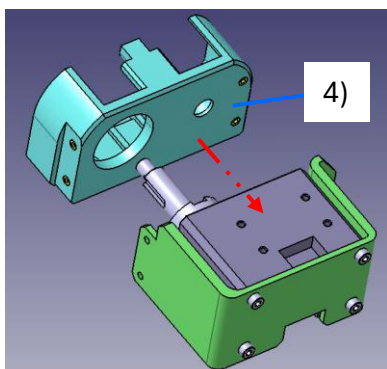
3) Secure BRACKET_DYNAMIXEL with CB2.5-15 (4 pcs). (Tighten lightly at this stage.)



- Install CASE

4) Install CASE onto DYNAMIXEL + BRACKET ASSY.

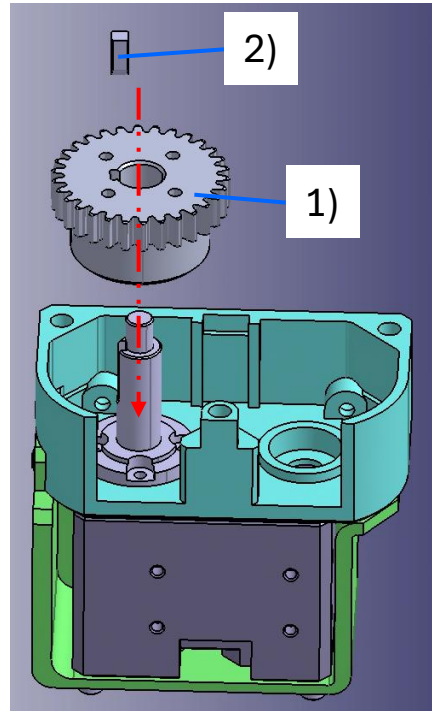
5) Secure BRACKET_DYNAMIXEL with CBSTNR2-5 (4 pcs). (Torque: 0.315 N·m)



(7) Installation of FINGER ASSY_R (Right Finger, Step (4))

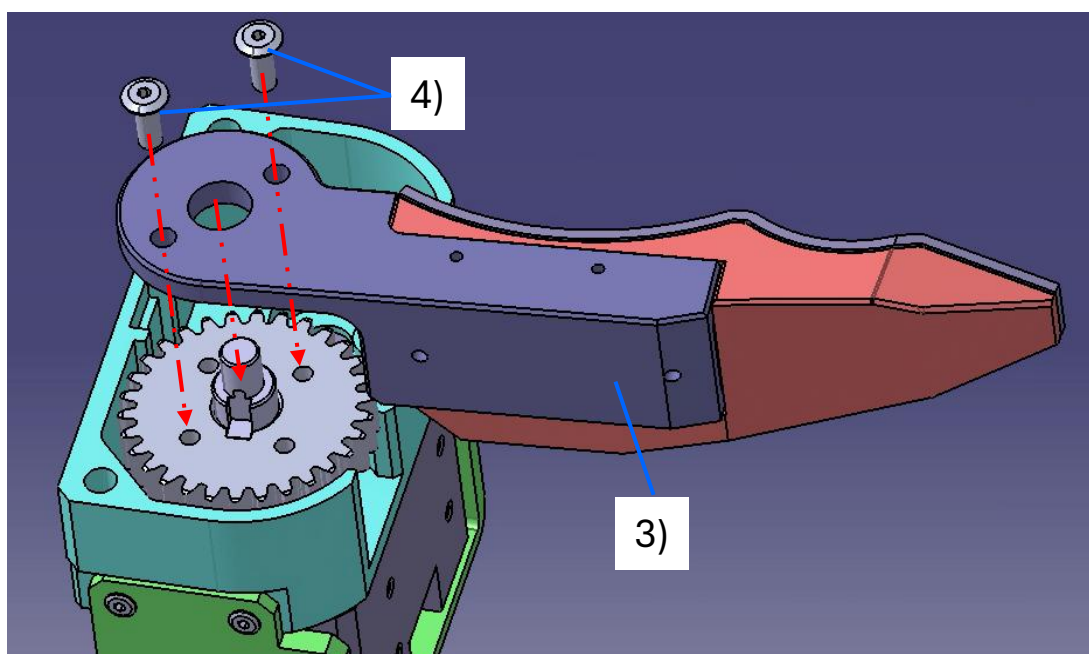
- Install SPUR GEAR

- 1) Install SPUR GEAR_z30 onto GEAR SHAFT.
- 2) Install PARALLEL KEY.



- Install FINGER ASSY_R

- 3) Install FINGER ASSY_R (4) onto GEAR SHAFT.
 - 4) Secure FINGER ASSY_R (4) with CBSTNR3-8 (2 pcs).
- (Apply threadlocker, Torque: 1.14 N·m)

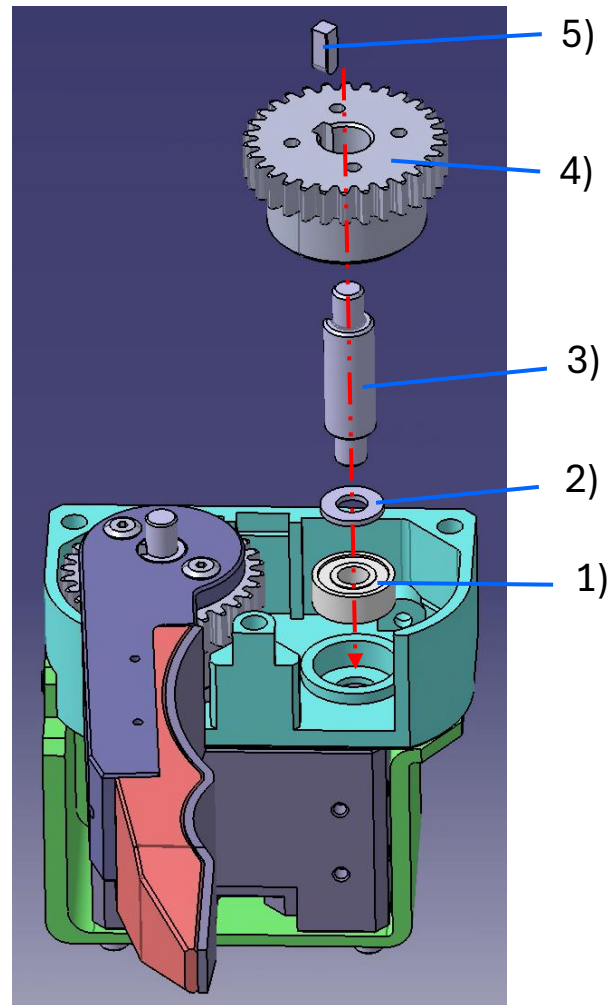


(8) Installation of FINGER ASSY_L (Left Finger, Step (5))

- Install SPUR GEAR

Install the following components onto the shaft of the left finger:

- 1) BEARING
- 2) WASHER
- 3) SHAFT_D8
- 4) SPUR GEAR_z30
- 5) PARALLEL KEY

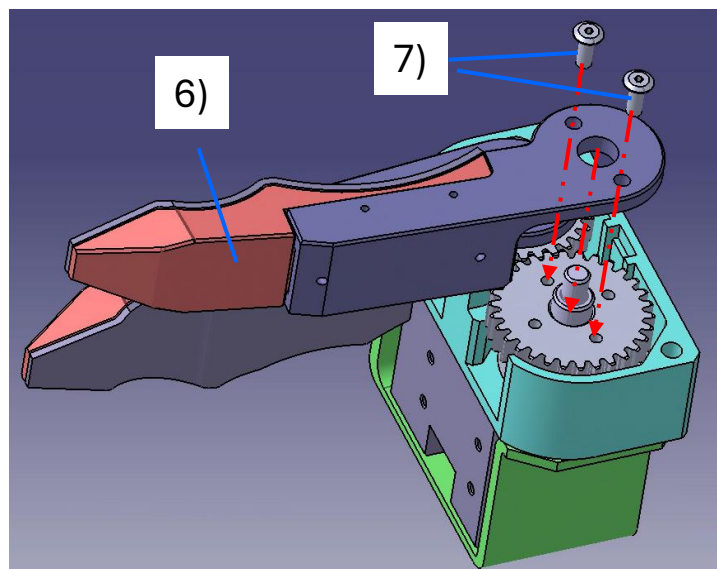


- Install FINGER ASSY_L

6) Install FINGER ASSY_L (5) onto SHAFT_D8.

7) Secure FINGER ASSY_L (5) with CBSTNR3-8 (2 pcs).

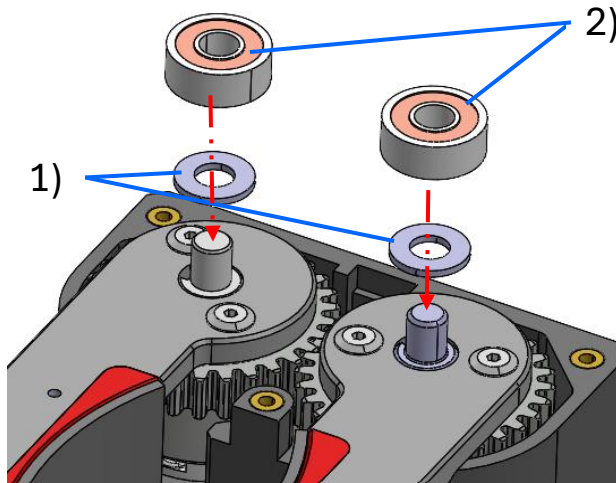
(Apply threadlocker, Torque: 1.14 N·m)



(9) Installation of UPPER PLATE

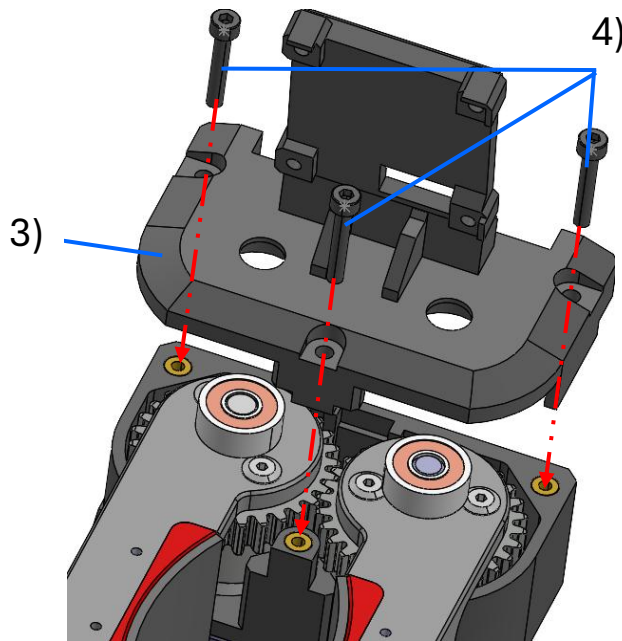
- Install bearings

- 1) Install washers onto the shaft sections of both left and right fingers.
- 2) Install bearings onto the shaft sections of both left and right fingers.

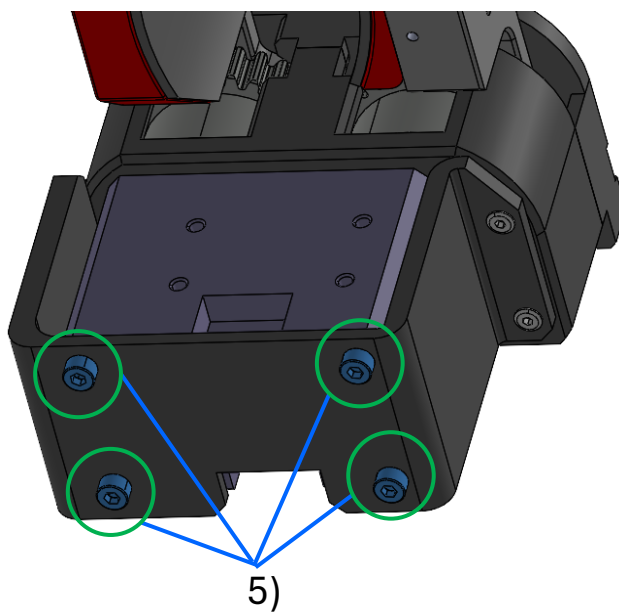


- Install UPPER PLATE

- 3) Install UPPER PLATE.
- 4) Secure UPPER PLATE with CB2.5-15 (3 pcs). (Torque: 0.65 N·m)



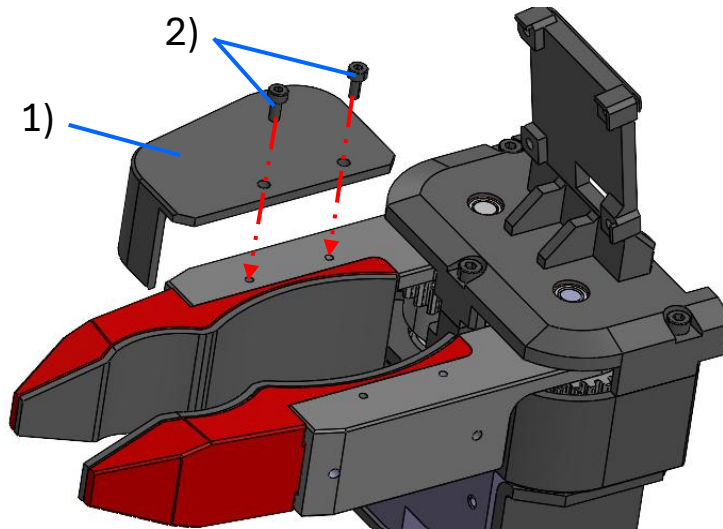
5) Tighten the DYNAMIXEL mounting bolts CB2.5-15 (4 pcs). (Torque: 0.65 N·m)



(10) Installation of FLAPs

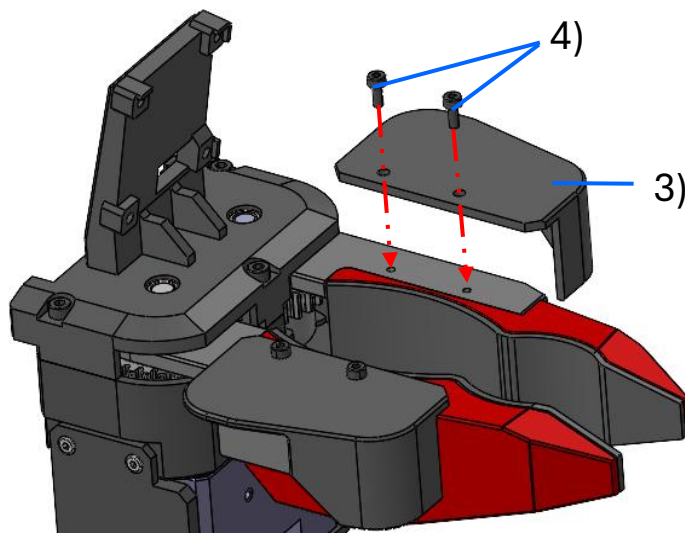
- Install FLAP_R

- 1) Install FLAP_R onto FINGER ATTACHMENT_R.
- 2) Secure FLAP_R with CB2-6 (2 pcs). (Torque: 0.315 N·m)



- Install FLAP_L

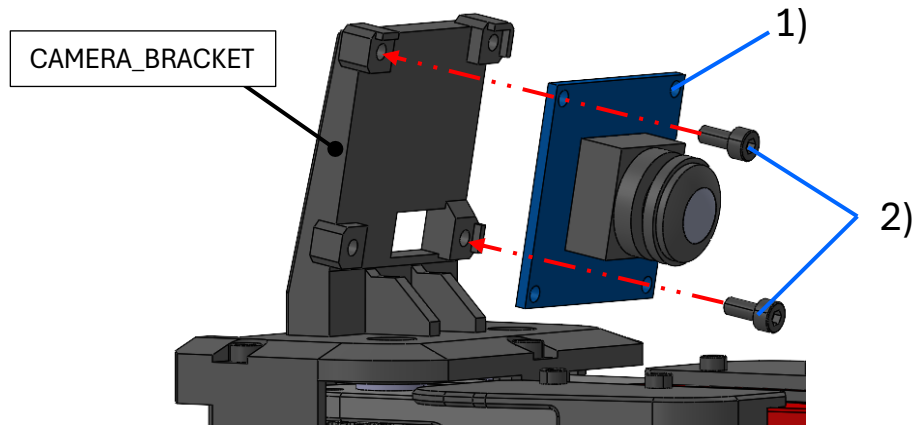
- 3) Install FLAP_L onto FINGER ATTACHMENT_L.
- 4) Secure FLAP_L with CB2-6 (2 pcs). (Torque: 0.315 N·m)



(11) Installation of CAMERA

- Install CAMERA

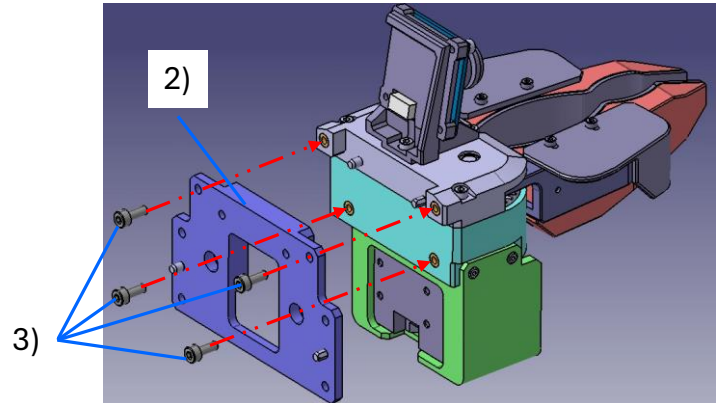
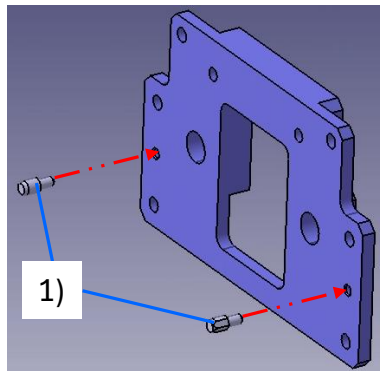
- 1) Place CAMERA onto CAMERA_BRACKET.
- 2) Secure CAMERA with CB2.5-6 (2 pcs). (Torque: 0.36 N•m)



(12) Installation of YUBI Gripper_DYNAMIXEL onto UR5e

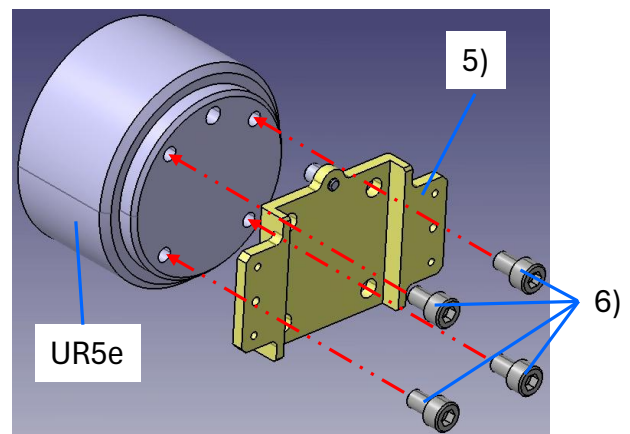
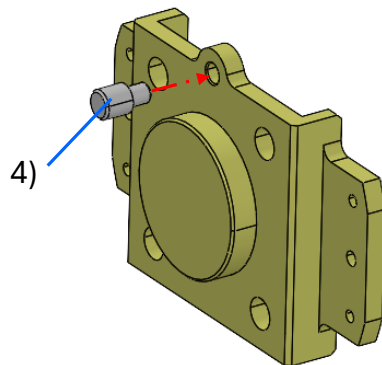
- Install BRACKET_GRIPPER

- 1) Install locating pins (JPBPB2-3: 1 pc / JPDPB2-3: 1 pc) onto BRACKET_GRIPPER.
- 2) Install BRACKET_GRIPPER onto the rear side of UPPER PLATE.
- 3) Secure BRACKET_GRIPPER with CBE3-8 (4 pcs). (Torque: 0.63 N·m)



- Install UR5e_FLANGE onto UR5e

- 4) Install locating pin (XDSHC6-P4-L6-B5: 1 pc) onto UR5e_FLANGE.
- 5) Install UR5e_FLANGE onto the UR5e tool flange.
- 6) Secure UR5e_FLANGE with CBE6-10 (4 pcs). (Torque: 9.2 N·m)



- Install YUBI Gripper_DYNAMIXEL onto UR5e

- 7) Align locating pins of BRACKET_GRIPPER with UR5e_FLANGE mounted on UR5e.
- 8) Secure with CBE3-8 (4 pcs). (Torque: 1.14 N·m)

